

Muzammal Naseer

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Education and Degrees

- 2018–2022 **Doctor of Philosophy in Engineering & Computer Science**, *Australian National University (ANU)*.
- Ph.D. Advisors: [Prof. Fatih Porikli](#), and [Dr. Salman Khan](#).
 - I defended my thesis titled *Novel Concepts and Designs for Adversarial Attacks and Defenses*.
 - My research was focused on understanding and improving out-of-distribution generalization of machine learning and deep learning based models. I developed new algorithms for adversarial attack and defense methods for computer vision applications including segmentation, object recognition and detection.
 - Multiple chapters of my Ph.D. thesis are also peer-reviewed and published at top machine learning and computer vision venues including NeurIPS, CVPR, ICCV, TNNLS, and TPAMI.
- 2011–2013 **Master of Science in Electrical Engineering**, *King Fahd University of Petroleum & Minerals (KFUPM)*, average GPA 3.75/4.0.
- I graduated with *first class honors* and defended my thesis titled *The Design of Finite Impulse Response (FIR) Wavefield Extrapolation Filters*.
 - Research was conducted with the collaboration of *Saudi Aramco* and the resulting software of my thesis was tested on the active oil well exploration sites. Multiple chapters of my thesis are also peer-reviewed and published at the Society of Exploration Geophysicists.
 - I took in-depth courses on mathematics, signal processing, systems identification, and control theory.
- 2006–2010 **Bachelor of Science in Electrical Engineering**, *University of the Punjab (PU)*, average GPA 3.89/4.0, Gold Medalist.
- I graduated by defending my thesis titled *Spectrum Monitoring & Analysis of Interference in EGSM Networks*. This project was funded by Pakistan Frequency Allocation Board (FAB) and telecommunication network operator company, ZONG.

Academic and Research Experience

- Sept. 2020–present **Postdoctoral Researcher**, *Computer Vision Laboratory, MBZUAI UAE*
- Currently, my research is focused on video understanding, adversarial learning, and developing insights by analyzing and visualizing the internal workings of deep neural networks.
 - I am supervising Master and Ph.D. students as well as research assistants.
 - I am actively participating in teaching including lab supervisions and guest lectures at MBZUAI.
- Sept. 2018 – Mar. 2020 **Research Associate**, *Inception Institute of Artificial Intelligence - IIAI, UAE*
- Developed cross-domain adversarial attack and task-generalizable neural purification defense.
- 2018 **Research Associate**, *Data61, CSIRO, Canberra, Australia*
- Developed task-generalizable transferable adversarial attack for neural networks vision systems.
- Sept. 2014 – June. 2017 **Lecturer (Full-Time)**, *University of Hafr Al Batin - subcampus of KFUPM, KSA*
- I joined subcampus of KFUPM, now known as University of Hafr Al-Batin as a full-time lecturer to teach variety of *Electrical Engineering* courses including circuit analysis, analog and digital systems, and programming (C/C++/Assembly). I also mentored student's final year graduate projects.
- 2013 **Teaching Assistant**, *Department of Electrical Engineering, KFUPM, KSA*
- Assisted in Control Theory and Signal Processing courses.
- 2009 **Research Assistant**, *Frequency Allocation Board (FAB), Pakistan*
- Worked with FAB team to develop the direction finding system for RF Interferers.

Honors and Awards

- 2019 Student travel award for Vancouver, Canada at *Neural Information Processing Systems* (NeurIPS).
- 2017 Postgraduate research scholarship by *ANU, Australia* for the period of three years.
- 2017 Fee remission merit scholarship by *ANU, Australia* for the period of four years.
- 2011 Master of science scholarship by the *Ministry of Higher Education, KSA* for the period of two years.
- 2011 Gold Medal by the *University of the Punjab* (PU) for outstanding performance in B.Sc. degree.
- 2007-09 University merit scholarship by the *University of the Punjab* (PU) consistently for three years.

Research Funding

- 2022-2025 **MBZUAI-WIS Program for Collaborative Research in AI, UAE**
 - **Title:** Biomimetic visual-recognition at low resolution with continuous learning.
 - **Amount:** 2.7 Million Dollar.
 - **Role:** Work Package Leader

Selected Publications

- CVPR 2022 **Self-Supervised Video Transformer.**
Oral, top 5.0% Kanchana Ranasinghe, **Muzammal Naseer**, Salman Khan, Fahad Shahbaz Khan, Michael Ryoo
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022.
- ICLR 2022 **On Improving Adversarial Transferability of Vision Transformers.**
Spotlight, top 5.0% **Muzammal Naseer**, Kanchana Ranasinghe, Salman Khan, Fahad Shahbaz Khan, Fatih Porikli
International Conference on Learning Representations (ICLR), 2022.
- NeurIPS 2021 **Intriguing Properties of Vision Transformers.**
Spotlight, top 3.0% **Muzammal Naseer**, Kanchana Ranasinghe, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, Ming-Hsuan Yang
Advances in Neural Information Processing Systems (NeurIPS), 2021.
- ICCV 2021 **On Generating Target Transferable Perturbations.**
Muzammal Naseer, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, Fatih Porikli
International Conference on Computer Vision (ICCV), 2021.
- ICCV 2021 **Orthogonal Projection Loss**
Kanchana Ranasinghe, **Muzammal Naseer**, Munawar Hayat, Salman Khan, Fahad Shahbaz Khan
International Conference on Computer Vision (ICCV), 2021.
- CVPR 2020 **A Self-supervised Approach for Adversarial Robustness**
Oral, top 5.7% **Muzammal Naseer**, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, Fatih Porikli
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2020
- NeurIPS 2019 **Cross-Domain Transferability of Adversarial Perturbations.**
Muzammal Naseer, Salman Khan, Harris Khan, Fahad Shahbaz Khan, Fatih Porikli
Advances in Neural Information Processing Systems (NeurIPS), 2019.

Selected Journal Publications

- TPAMI, 2022 **Stylized Adversarial Defense**
Muzammal Naseer, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, Fatih Porikli
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2022. [Impact factor: 16.39]
- TNNLS 2022 **Guidance Through Surrogate: Towards a Generic Diagnostic Attack**
Muzammal Naseer, Salman Khan, Fatih Porikli, Fahad Shahbaz Khan
IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2022 [Impact factor: 10.45]

Selected Other Conference Publications

- BMVC, 2021 **Rich Semantics Improve Few-shot Learning**
Mohamed Afham, Salman Khan, Haris Khan, **Muzammal Naseer**, Fahad Khan
The British Machine Vision Conference (BMVC), 2021
- WACV 2019 **Local Gradients Smoothing: Defense Against Localized Adversarial Attacks**
Muzammal Naseer, Salman Khan, Fatih Porikli
Winter Conference on Applications of Computer Vision (WACV), 2019

Computational Skills

- Python I am extensively using python to build machine learning algorithms for the last few years.
- Pytorch It is usually my default choice due to its dynamic nature and object-oriented graph design approach.
- Tensorflow I have used Tensorflow to build different projects including a defense algorithm against unconstrained adversarial attacks.
- C/C++ I scored A+ grades in these languages in my B.Sc courses. [[Transcript](#)]

Research Supervision

- 2021–present Hasmat Shadab Malik, MS student at MBZ University of AI (MBZUAI, UAE)
- 2021–present Shahina Kunhimon, Ph.D. student at MBZ University of AI (MBZUAI, UAE)
- 2020–2021 [Kanchana Ranasinghe](#), Research associate at MBZ University of AI (MBZUAI, UAE)
- 2020–2021 [Mohamed Afham](#), Research associate at MBZ University of AI (MBZUAI, UAE)

Professional Services

Program Committees

- 2022 [Vision Transformers: Theory and applications](#), ACCV 2022
- 2021 [Adversarial Robustness In the Real World](#), ICCV 2021
- 2020 [Adversarial Robustness In the Real World](#), ECCV 2020

Reviewer

- From 2022 Int. Conf. Computer Vision and Pattern Recognition (CVPR)
- From 2022 Int. Conf. Advances in Neural Information Processing Systems (NeurIPS)
- From 2021 Int. Journals: TPAMI, TIP, and TNNLS

References

- Dr. Salman Khan**, Associate Professor at MBZ University of AI (MBZUAI, UAE)
✉ salman.khan@mbzuai.ac.ae
- Dr. Munawar Hayat**, Senior Lecturer at Monash University
✉ munawar.hayat@monash.edu.au
- Dr. Fahad Shahbaz Khan**, Associate Professor at MBZ University of AI (MBZUAI, UAE)
✉ fahad.khan@mbzuai.ac.ae,
- Prof. Fatih Porikli**, Chief Scientist at Qualcomm, United States
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